

CSMFAB000000000113-Vol-26

Cost Analysis: AMSR-UBSCLS Across All Scales

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1. Bill of Materials Costs by Scale

Scale 1 — Nano Hobbyist (50 g spear)

Component	Material	Qty	Unit Cost	Total
Spear body	ZrB2-SiC, in-house SHS	0.06 kg	\$102/kg	\$6
Screw blades	ZrB2-SiC thin cast	0.01 kg	\$150/kg	\$2
UHMWPE fiber (ropes)	Dyneema SK75	40 m	\$0.25/m	\$10
Smart Rope segments	ZrB2-SiC cubes (16 ea × 2 ropes)	32	\$0.87	\$28
KNbO3-BaTiO3 actuators	TDK 2025	32	\$0.45	\$14
LoRa modules	RAK3172	34	\$6.50	\$221
BFRP vessel reinforcement	Elium@+basalt	2 kg	\$25/kg	\$50
Instrumentation (basic)	MEMS + LoRa + MRAM	1 set	\$180	\$180
Pneumatic spin-up	BFRP N2 cylinder	1	\$45	\$45
Safety/misc	—	—	—	\$50
TOTAL SYSTEM				\$606

Vessel charter cost (kayaks): \$0 (hobby items) Per-launch amortized: \$200 (rope/hardware reusable ×3 launches)

Total cost per launch: ~\$200-\$606 (hobbyist nano scale)

Scale 2 — Micro Amateur (1 kg spear)

Component	Cost
Spear body (ZrB2-SiC)	\$50
Smart Rope set (4 × 15m)	\$380
Instrumentation (basic)	\$320
Vessels (small RIBs, charter)	\$800
Labor/logistics	\$400
TOTAL	\$1,950 per launch

Scale 3 — Small Amateur (10 kg spear)

Component	Cost
Spear body (ZrB2-SiC)	\$400
SRM second stage (GIC-immune)	\$2,800
Smart Rope set (6 × 30m)	\$3,200
Instrumentation (full)	\$1,200
Vessels (charter 6 boats 1 day)	\$4,500

Component	Cost
Dive team/ROV	\$2,000
Safety/logistics	\$1,500
FAA NOTAM filing fee	\$0 (voluntary)
TOTAL	\$15,600 per launch

Scale 4 — Medium (100 kg spear)

Estimated: \$85,000-\$120,000 per launch (institutional-grade; comparable to small sounding rocket)

Scale 5 — Large (1,000 kg spear)

Estimated: \$450,000-\$750,000 per launch (comparable to small orbital launch attempt)

2. Reuse and Amortization

Smart Rope lifetime: 3-10 launches (limited by UHMWPE UV degradation for surface ropes) Spear lifetime: 1 launch (ZrB₂-SiC — expendable for high-velocity flights; recoverable at nano/micro scale) Vessels: chartered per launch or owned (1-3 year payback for frequent use)

At 10 launches/year, Scale 3: - Fixed costs (vessels, equipment): \$30,000/yr - Variable per launch: \$8,000 - Total annualized: \$110,000/yr = \$11,000 per launch

\$11,000 per suborbital research launch — competitive with sounding rockets (\$40,000-\$150,000).

Citations (Vol 26)

- CSMFAB Materials Study Part VI, in-house synthesis costs, CSM 2026
- SpaceX Falcon 1 historical costs (reference for small orbital launch economics)
- NASA Sounding Rocket Program, cost per mission data, 2024
- DSM Dyneema SK75/SK99 commercial pricing, 2025

End Vol 26 / CSMFAB000000000113 / CSM Safe Pod Engineering Company / June 2026